O-R-500D
January 31, 1978
SUPERSEDING
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August 18, 1970

FEDERAL SPECIFICATION

RODENTICIDE, BAIT, ANTICOAGULANT

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 Scope. This specification covers a rodenticide consisting of a bait and an anticoagulant. The anticoagulant may be diphacinone, chlorophacinone, fumarin, pivalyl, or warfarin as specified in 3.1.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specification:

PPP-C-96 - Cans, Metal, 28 Gage and Lighter.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Philadelphia, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Houston, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 6840

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Laws and Regulations:

40 CFR 162-180 - Regulations for the Enforcement of the Federal Insecticide, Fungicide, and Rodenticide Act.
49 CFR 170-189 - Hazardous Materials Regulations.

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Composition. The rodenticide shall consist of a chemical anticoagulant which may be chlorophacinone, diphacinone, fumarin, pivalyl, or warfarin suitably compounded in a bait meeting requirements of Table I and 3.4. The bait shall be free-flowing pellets.

TABLE I. Alternative anticoagulant requirements (Option as specified; see 6.2(d))

Ingredient	Percent minimum	Percent maximum	Test method paragraph
Chlorophacinone	0.005	0.0055	4.4.2.1
Diphacinone	.005	.0055	4.4.2.2
Fumarin	.025	.028	4.4.2.3
Pivalyl	.025	.028	4.4.2.4
Warfarin	.025	.028	4.4.2.5

3.2 Particle size. Not more than 30 percent of the finished rodenticide shall pass through a No. 20 U.S. Standard woven-wire sieve, when tested as described in 4.4.3.

3.3 Moisture. The rodenticide shall not lose more than 12 percent of its weight when tested as described in 4.4.4.

3.4 Registration and labeling. The rodenticide shall be registered and labeled in compliance with the Federal Insecticide, Fungicide, and Rodenticide Act (FIFRA), as amended. This act, and regulations promulgated thereunder, is administered by the Environmental Protection Agency (EPA). The containers shall be labeled in accordance with 49 CFR 170-178, with contents and directions for use.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government.

4.2 Examination of preparation for delivery. An examination shall be performed to determine compliance with the packaging, packing, unitization, and marking requirements of section 5. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for examination at one time. The packaging and packing shall be examined for the defects specified in table II. The inspection level shall be S-2 with an AQL of 4.0 expressed in terms of percent defective.

TABLE II. Examination of preparation for delivery

Examine	Defect
Container	Not as specified.
Content	Not as specified.
Marking	Omitted; incorrect; illegible; improper size, location, sequence or method of application.
Material	Component missing or damaged.
Workmanship	Bulging or distortion of containers.

4.3 Examination of contents of unit container. The sample unit shall be one filled container. Sampling shall be in accordance with MIL-STD-105 at inspection level S-2 and an acceptable quality level (AQL) of 4.0 percent defective. The examination shall be made to determine compliance with 3.1, 3.4, and 5.3. The sample units shall be examined for defects as listed in table III.

TABLE III. Examination of unit container

Examine	Defects
Markings	Not as specified (see 3.4 and 5.3).
Contents	Quantities not as specified.

4.4 Testing.

4.4.1 Sampling for testing. The sample unit shall be defined as one filled unit container. A 1-pound composite shall be made by mixing equal portions of each of the sample units taken from the lot in accordance with inspection level S-1. The lot shall be unacceptable if the composite sample fails to meet any of the requirements described in section 3.

4.4.2 Testing procedure. Test for the active ingredient specified in the procurement document (see 6.2d).

4.4.2.1 Determination of chlorophacinone in bait. Grind approximately 21 g of bait to a fine powder. Weigh to the nearest mg 20 g of the powder; place it into the bag of the Soxhlet extractor and extract with 250 ml of ethyl acetate for 3 hours. Allow the extract to cool. Place inside the neck of a 250-ml decanting flask some glass fiber and 6 g of WOELM basic alumina. Filter the preceding extract and reject the filtered liquid. Wash the alumina column with 5 fractions of 10 ml of ethyl acetate, then with 10 ml of ethanol at 96°C ± 0.5°C. Reject the wash solvents. Close the top of the decanting flask and extract the chlorophacinone

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adhered to the alumina twice with 50 ml of ammoniated ethanol (10 ml aqua ammonia plus 90 ml of ethanol at 96°C). This extraction is made by mechanical agitation for 10-15 minutes. Collect the extracts in a balloon of a rotating evaporator (the extracts are filtered by the glass fiber inserted in the neck of the decanting flask). Rinse the decanting flask with 20 ml of ammoniated ethanol and combine this rinse solvent with the preceding extracts. Concentrate the ammoniated ethanol solutions to a volume of 4 to 5 ml in vacuum at a temperature of 50° to 60° C. Collect the concentrated solution in a test tube graduated at 10 ml, rinse the balloon of the evaporator with a few ml of ammoniated ethanol (10 ml aqua ammonia + 90 ml ethanol at 96°C), and add this rinsing solvent to the preceding concentrated solution. Fill up to a volume of 10 ml with the ammoniated ethanol. Homogenize the solution. Filter the solution by means of a filter paper of slow filtration. Trace the spectra of UV absorption from 200 to 350 nm of the test solutions. Measure the optical density at 326 nm.

Calculations:

The amount of chlorophacinone is then found by the following formula:

$$\frac{(100)\ (a)\ x\ (10)}{(b)\ (\text{Sample Weight})} = \%\ \text{chlorophacinone}$$

where (a) = the optical density of the sample.
(b) = the optical density of a std sample of chlorophacinone.

4.4.2.2 Determination of diphacinone in bait. Weigh to the nearest mg a 20.0 gm sample into a Soxhlet thimble and extract with ethyl ether* for 4 hours. Evaporate the extract to less than 50 ml on a steam bath, transfer to a 50 ml volumetric flask, and make to volume with ethyl ether*. Pipet 2 ml of this solution into a glass stoppered 16 x 150 mm test tube and 2 ml ethyl ether* into another test tube to serve as instrument blank. To each tube add 10 ml freshly made 1 percent pyrophosphate solution by pipet (1 percent sodium pyrophosphate solution - dissolve 5 g $Na_4P_2O_7.10H_2O$ in 500 ml water). Shake vigorously for 2 minutes. Centrifuge at high speed until the aqueous layer is clear. Draw off the ether layer by means of an aspirator connected to a glass tube drawn into a fine tip. Add 2 ml eethyl ether*, shake vigorously, centrifuge, and draw off the ether layer. Repeat with another 2 ml of ethyl ether. Finally repeat twice with 2 ml portions petroleum ether solvent (petroleum ether solvent - extract 200 ml petroleum ether with 3 portions of 20 ml sodium pyrophosphate solution). Determine the absorbance of the sample solution in a suitable spectrophotometer (Beckman model DU or equivalent) at 286 nm in 1 cm silica cuvettes against the blank solution. Calculate the percent diphacinone from:

$$\%\ \text{diphacinone} = \frac{\text{Absorbance X } 0.226}{\text{Sample Weight}}$$

This method is designed for products containing about 0.005 percent diphacinone (2-diphenylacetyl-1,3-indandione). The principal peak for diphacinone is at 286 nm and is rather near the 283 nm peak for pivalyl for which the absorption curves are similar. Since these peaks are rather strong, a maximum at 286 nm indicates that diphacinone rather than pivalyl is present.

4.4.2.3 Fumarin determination in bait. Weigh to the nearest mg 10.0 gm of sample into a 250 ml glass stoppered flask and add 100 ml of ethylene dichloride with the pipette. Shake on a shaking machine for at least 10 minutes. Decant liquid to a centrifuge tube, stopper, and centrifuge 5 minutes, or until clear. Pipette 2 ml of the ethylene dichloride extract into a glass stoppered cylinder, add 10 ml of a 1 percent NaOH solution with a pipette, and shake for 1 minute by hand. Prepare a blank solution in the same manner, using 2 ml ethylene dichloride in place of the extract. Decant the sodium hydroxide layer into a centrifuge tube and centrifuge until clear. Pipette a sufficient amount of the alkali layer (approximately 3 ml) into a quartz cuvette (1 cm cell) and determine the optical density at 305 nm and slit opening of about 0.86 nm, using a Bechmann spectrophotometer, Model DU or equivalent.

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CALCULATION

The optical density of 1 mg of fumarin in 100 ml of 1 percent NaOH solution was found to be 0.458, using the above procedure. If A is the optical density obtained:

$$\% \text{ fumarin} = \frac{A \ (10/100) \ (50) \ (100)}{0.458 \ (1000) \ (\text{Sample Weight})} = \frac{A \ (1.0917)}{\text{Sample Weight}}$$

4.4.2.4 Pivalyl determination in bait:

REAGENTS

(a) Sodium pyrophosphate solution - dissolve 5 gm $Na_4P_2O_7.10H_2O$ in 500 ml water.
(b) Ethyl ether* - n-hexane solvent (20-80) - extract 200 ml n-hexane with three 20 ml portions of pyrophosphate solution (a) and mix the extracted n-hexane with 50 ml ethyl ether.
(c) Hydrochloric acid - mix 4 parts water with 1 part conc. HCl.

PREPARATION OF SAMPLE:

Weigh to the nearest mg 2.0 g ground sample into a 125 ml glass stoppered flask and add 50 ml sodium pyrophosphate solution (a) with a pipette. Shake for 1 to 2 hours on a shaking machine. Transfer 15 ml to a glass stoppered test tube and centrifuge for at least 5 minutes.

Pipette 10 ml of the solution into a centrifuge tube, and in another centrifuge tube pipette 10 ml of the pyrophosphate solution (a) to be used as a blank. To both tubes add 3 ml HCl and 25 ml ether-hexane solvent (b) and shake for 10 minutes. Centrifuge if necessary.

Pipet 5 ml of the solvent layer from each tube into clean glass stoppered test tube and pipette 5 ml pyrophosphate (a) and shake for two minutes. Centrifuge and remove the solvent layers with the aid of an aspirator connected to a glass tube drawn to a fine tip.

Determine the absorbance of the sample solution in a suitable spectrophotometer (Beckman model DU or equivalent) at 283 nm in 1-cm silica cuvettes against the treated blank solution. Calculate the percent pivalyl from the following:

$$\% \text{ pivalyl} = \frac{\text{Absorbance X } 0.114}{\text{Sample Weight}}$$

NOTE:

(1) With a model DU the solution can be scanned from 360 to 250 nm to identify the pivalyl.

4.4.2.5 Warfarin determination in bait. Weigh to the nearest mg 10.0 of sample into a 100 ml centrifuge tube. Add 50 ml of ethyl ether*, stopper tightly, and shake on a mechanical shaking machine for about 30 minutes. Place the tube in a centrifuge and spin until the ether layer is clear. Suspend 2 grams of Attapulgus Clay** in ethyl ether* and transfer to a chromatographic tube, fitted with a capillary stopcock. (A plug of glass wool holds the absorbent in place). Using air pressure, force the excess ether through the tube until the absorbent bed is just covered. Carefully add a 2-ml aliquot of the centrifuged ether layer, force the ether solution into the bed, and fill the tube with ether. Apply air pressure to the top of the column so that the ether passes out of the column at about two drops per second. Collect 25 ml of the eluate in a 50 ml glass stoppered mixing cylinder. Add 10 ml of 1 percent sodium pyrophosphate solution and shake for 2 minutes. Place the cylinder in a centrifuge and spin at high speed until the lower aqueous layer is clear. Draw off the upper ether layer, including any emulsion that persists at the interface, by means of an aspirator connected to a tube drawn to a fine tip. Add about 2 ml ethyl ether to the test tube, shake vigorously, centrifuge, and completely draw off the ether layer. Repeat with a second ether extraction, and then extract twice with petroleum ether (which has previously been extracted with 1 percent sodium pyrophosphate solution). Completely draw off the solvent layer and allow air to be drawn over the liquid surface for a few seconds.

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Fill a 1 cm quartz cuvette with the aqueous solution and determine the maximum optical density in the range 305 - 310 nm using a Beckman Model DU spectrophotometer, using for the zero setting a 1 percent sodium pyrophosphate solution which has been extracted with ethyl ether and petroleum ether (previously treated with 1 percent sodium pyrophosphate solution).

% warfarin = (A/A') x (0.250/g sample)

where A = optical density of sample
A' = optical density of std warfarin soln. (available from WARF Institute, P.O. Box 2037, Madison, WI 53701).

* Use U.S.P. Grade or add corresponding amount of ethyl alcohol to A. R. Absolute.
** Attapulgus clay, grade 200/up or Attaclay, Attapulgus Clay Co., 210 W. Washington Square, Philadelphia 5, PA. Both clays should be first tested with a standard warfarin solution in ether prior to use to determine if the warfarin can be completely recovered.

4.4.3 Sieve test. Transfer 100 ± 1 grams of the rodenticide to a dried No. 20 (840 micron) U.S. Standard sieve and screen in a mechanically operated sieve for 10 minutes. Weigh the material which passes through the sieve and calculate the percentage to the nearest tenth of a percent.

4.4.4 Moisture. Weigh to the nearest mg 20 g of the rodenticide into a tared container and dry for 10 hours under atmospheric pressure at a temperature of 105°C (± 2°C); cool in desiccator and weigh. Dry again for one hour, or until change in weight is not more than 2 mg. Report loss in weight as moisture.

5. PREPARATION FOR DELIVERY

5.1 Packaging shall be level A or Commercial, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit packaging. The rodenticide shall be packaged in quantities of 2 ounces, 1 pound, 1-1/2 pounds (24 ounces), 3 pounds, 5 pounds, or 6 pounds, as specified (see 6.2), in round, hermetically sealed, open-top cans conforming to PPP-C-96, type I, plan B coating. The product shall be compressed to a density of at least 1 ounce for each 2.6 cubic inches based on the net inside packing space of the can.

5.1.1.2 Intermediate packaging. Intermediate packaging of 2-ounce cans shall be as specified in the appendix to PPP-C-96 for level A packaging.

5.1.2 Commercial. The rodenticide shall be preserved and packaged in accordance with normal commercial practice. The complete package shall be designed to protect the item against damage during shipment, handling, and storage. Quantity of product per package shall be as specified in the procurement document (see 6.2).

5.2 Packing. Packing shall be level A or Commercial, as specified (see 6.2).

5.2.1 Level A. Packing shall be in accordance with the appendix to PPP-C-96 for level A packing. Twenty-four 1-1/2 pound cans shall be packed in the shipping container. The 6-pound cans shall be packed in the shipping container in the same arrangement designed for the 5-pound cans. Four intermediate packages of 2-ounce cans shall be packed in the shipping container.

5.2.2 Commercial. The rodenticide shall be packed in shipping containers to insure safe delivery at destination, to provide for safe redistribution by the initial receiving activity, and shall be acceptable by common carrier under the National Motor Freight Classification or Uniform Classification.

5.3 Marking.

5.3.1 Marking shall be as specified in the contract or order.

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5.3.2 Labeling. The rodenticide shall be registered and labeled in compliance with the Federal Insecticide, Fungicide, and Rodenticide Act and the Federal Hazardous Materials Regulations (49 CFR 170-189).

5.4 Unitization. When shipments to Government depots are full car or truckload, the shipping containers shall be unitized for shipment in accordance with normal commercial practice. The unitized load shall not exceed 2,500 pounds in weight, 63 inches in height, 56 inches in length, and 45 inches in width.

6. NOTES

6.1 Intended use. Rodenticide is intended for use in the control of rats and mice.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Packaging and packing required (see 5.1 and 5.2).
(c) Quantity to the unit package (see 5.1.1.1.
(d) Whether diphacinone, fumarin, pivalyl, chlorophacinone, or warfarin is required (see 3.1).

6.3 The Federal Insecticide, Fungicide, and Rodenticide Act of 1947 is under the jurisdiction of the Pesticides Regulation Division, Agricultural Research Service, U.S. Department of Agriculture, Washington, D.C.

6.4 Chemical description of anticoagulant. The chlorophacinone shall be the chemical 2-[(p-chlorophenyl) phenylecetyl]-1,3-indandione. The diphacinone shall be the chemical 2-(diphenylacetyl)-1,3-indandione. The fumarin shall be the chemical 3-(alpha-acetonyl-furfuryl)-4-hydroxycourmarin. The pivalyl shall be the chemical 2-pivalyl-1,3-indandione. The warfarin shall be the chemical 3-(alpha-acetonylbenzyl)-4-hydroxycoumarin.

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